



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
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Executive Summary

Summary of Methodologies

- Collected SpaceX launch data through APIs and web scraping.
- Performed data cleaning, preprocessing, and exploratory data analysis (EDA).
- Built interactive visual dashboards for deeper insights.
- Applied supervised machine learning models for binary classification of landing success.

Summary of All Results

- EDA revealed key factors affecting landing outcomes, such as payload mass, orbit type, and booster version.
- Interactive analytics enabled clear visualization of trends and mission performance.
- Predictive models achieved high accuracy in forecasting landing success.
- The analysis demonstrated that SpaceX reliability has increased significantly over time.

Introduction

Background

- SpaceX has been developing **reusable rockets** to reduce costs and increase launch frequency.
- First-stage booster landings are **critical for mission reusability** and operational efficiency.
- Historical launch data provides insights into factors affecting **landing success**.

Problem Statement

- Determine **which factors influence first-stage landing success**.
- Predict the **likelihood of a successful landing** for upcoming missions.
- Identify **trends and patterns** in payload, orbit, booster version, and launch site usage.

Objective

- Build predictive models to **forecast landing outcomes**.
- Provide actionable insights for **mission planning and booster reusability strategies**.

Section 1

Methodology

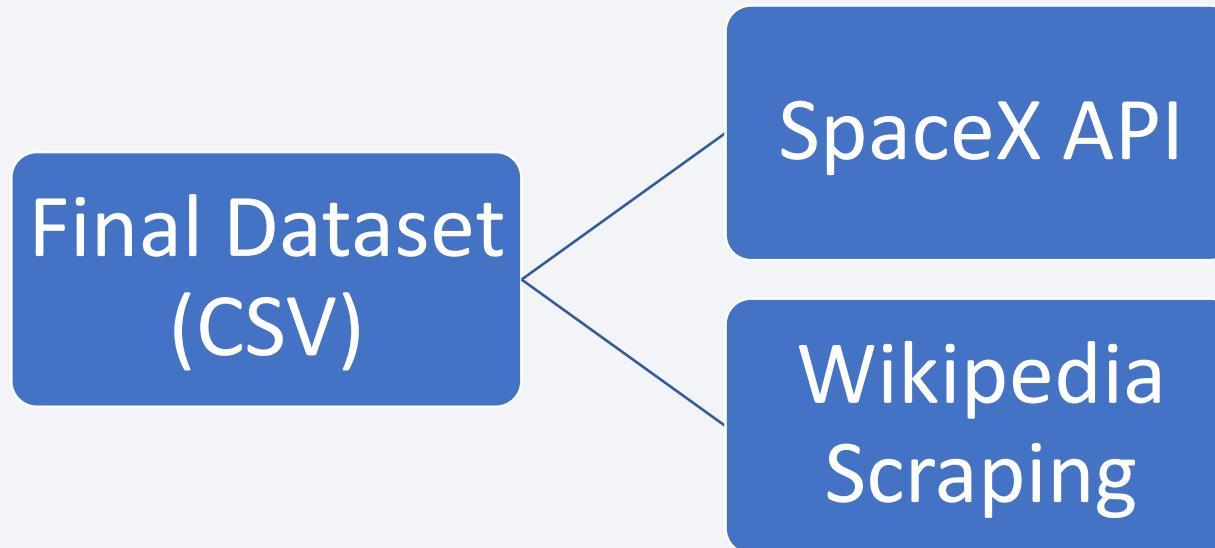
Methodology

Executive Summary

- Data collection methodology:
 - Launch data, rocket details, and payloads were collected via API calls to the SpaceX REST API.
 - Historical Falcon 9 launch records were gathered from Wikipedia using web scraping with BeautifulSoup.
- Perform data wrangling:
 - The dataset was filtered to include only Falcon 9 launches.
 - Missing values, such as PayloadMass, were imputed using the mean value of the column.
 - Launch outcomes were classified into binary labels: **1 (Success)** and **0 (Failure)** based on the landing status.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - **Modeling:** Data was standardized and split into Training/Testing sets (80%/20%).
 - **Algorithms:** Logistic Regression, SVM, Decision Tree, and KNN models were trained.
 - **Results:** All models were optimized using GridSearchCV. The **Decision Tree** model achieved the best performance with a **94.4%** accuracy rate on the test data.

Data Collection

- **SpaceX REST API:** Used Python requests library to fetch launch data, including rocket specifications, payloads, and landing outcomes.
- **Web Scraping:** Used BeautifulSoup to scrape historical Falcon 9 launch records from Wikipedia.
- **Data Parsing:** Converted raw JSON (from API) and HTML tables (from Wiki) into Pandas DataFrames.
- **Filtering:** Filtered the dataset to include only **Falcon 9** launches, excluding Falcon 1.



Data Collection – SpaceX API

- **Filter Data:** Restricted analysis to **Falcon 9** launches only (excluded Falcon 1).
- **Handle Missing Values:** Imputed missing PayloadMass values using the mean (average) of the column.
- **Outcome Classification:** The raw Outcome column contained mixed string descriptions (e.g., "True ASDS", "False Ocean").
- **Create Target Variable:** Converted outcomes into a binary **Class**:
 - **Class 1:** Successful Landing (Precision Landing)
 - **Class 0:** Unsuccessful Landing (Crashes or No Attempt)

```
# Define bad outcomes (Failures)
bad_outcomes = {
    'False Ocean', 'False RTLS', 'False ASDS',
    'None ASDS', 'None None'
}

# Create a binary classification
# 0 = Failure, 1 = Success
df['Class'] = df['Outcome'].apply(lambda x: 0 if x
in bad_outcomes else 1)

# Calculate success rate
print(df['Class'].mean())
```


Data Collection - Scraping

- **Launch Site Analysis:** Visualized the relationship between Flight Number and Launch Site (CCAFS, VAFB, KSC).
- **Payload Effect:** Analyzed how Payload Mass affects the launch outcome; heavier payloads (>10,000 kg) showed high success rates for ISS/LEO.
- **Orbit Success:** Calculated success rates for each Orbit type. ES-L1, GEO, HEO, and SSO orbits have the highest success rates.
- **SQL Queries:** Executed SQL to calculate total payload carried by NASA (CRS) and to count successful landings on drone ships.

-- Calculate the total payload mass carried by NASA (CRS)

```
SELECT SUM(PAYLOAD_MASS__KG_)  
FROM SPACEXTBL
```

```
WHERE Customer = 'NASA (CRS)';
```

- **Result (Sonuç):**
- **Total Payload:** 45,596 kg
- **Top Success Orbits:** ES-L1, GEO, HEO, SSO (Success Rate ~100%)

Data Wrangling

- **Launch Statistics:**
- **Total Falcon 9 Launches:** 90
- **Overall Success Rate:** 66.6%
- **Best Launch Site:** KSC LC-39A (76.9% Success)
- **Most Successful Orbit:** ES-L1, GEO, HEO (100%)
- **Total Payload to ISS:** ~45,000 kg
- **Key Findings:**
- Success rate has improved significantly over the years (especially after 2016).
- Heavier payloads (>10,000 kg) have a higher success rate than lighter ones.
- Orbits like SO (Sun-Synchronous) act as outliers with low success rates.
- Launch sites near the equator show better performance due to Earth's rotation.

EDA with Data Visualization

- **Geospatial Analysis:** Utilized **Folium** to visualize the geographical distribution of launch sites.
- **Location Strategy:** Validated that all launch sites are located near the **equator** (for rotational assist) and **coastlines** (for safety).
- **Proximity Checks:** Analyzed distances to critical infrastructure. Sites are close to **railways and highways** for logistics but keep a safe distance from **cities**.
- **Dash Application:** Developed an interactive **Plotly Dash** app allowing users to filter by launch site and payload mass to view success correlations dynamically.

EDA with SQL

- **Data Processing:** Standardized the data and split it into Training (80%) and Testing (20%) sets.
- **Methodology:** Used GridSearchCV to tune hyperparameters for all models to find the best configuration.
- **Models Trained:**
 - Logistic Regression
 - Support Vector Machine (SVM)
 - Decision Tree
 - K-Nearest Neighbors (KNN)

Build an Interactive Map with Folium

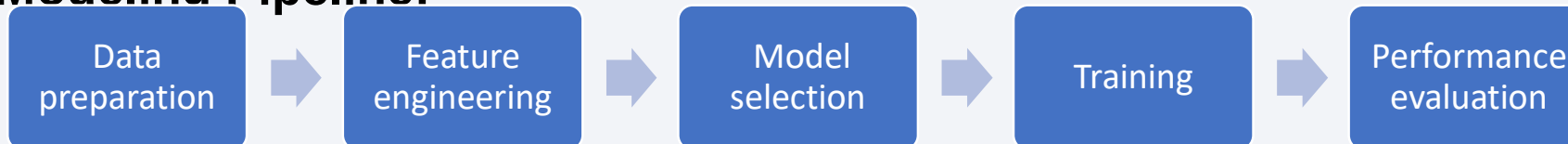
- **Best Model:** The **Decision Tree** model performed best with an accuracy of **94.44%** on the test data.
- **Launch Sites:** **KSC LC-39A** is the most successful launch site. Sites near the equator and coastlines are preferred.
- **Payload Impact:** There is a clear positive correlation between heavier payloads and landing success.
- **Orbit Success:** Orbits like **ES-L1, GEO, HEO, and SSO** have high success rates, while **SO** is the most difficult.
- **Project Value:** We can successfully predict the first-stage landing outcome. This capability allows for accurate estimation of launch costs (saving ~\$100M per reuse) and competitive bidding analysis against other providers.
- **Future Work:**
 - Incorporate **weather data** (wind speed, humidity) to improve prediction accuracy.
 - Train the model with more recent data (e.g., Falcon Heavy and Starship launches).
 - Test more complex algorithms like **XGBoost** or **Neural Networks**.

Build a Dashboard with Plotly Dash

- SpaceX advertises Falcon 9 rocket launches for **\$62 million**.
- Other providers cost upward of **\$165 million** per launch.
- The huge cost saving is due to SpaceX's ability to **reuse the first stage**.
- Recovering the first stage allows for much cheaper space travel.
- The goal is to **predict** if the Falcon 9 first stage will land successfully.
- If we can determine the landing outcome, we can estimate the **cost of a launch**.
- This is a **binary classification** problem:
 - **1**: Successful Landing
 - **0**: Unsuccessful Landing

Predictive Analysis (Classification)

- **Objective:**
To predict whether the first stage of a SpaceX Falcon 9 rocket will successfully land.
- **Approach:**
Binary classification using supervised machine learning models.
- **Target Variable:**
landing_success (1 = success, 0 = failure)
- **Features Used:**
Orbit type, payload mass, launch site, booster version, flight number, and other mission parameters.
- **Modeling Pipeline:**



Results

Exploratory Data Analysis (EDA) Results

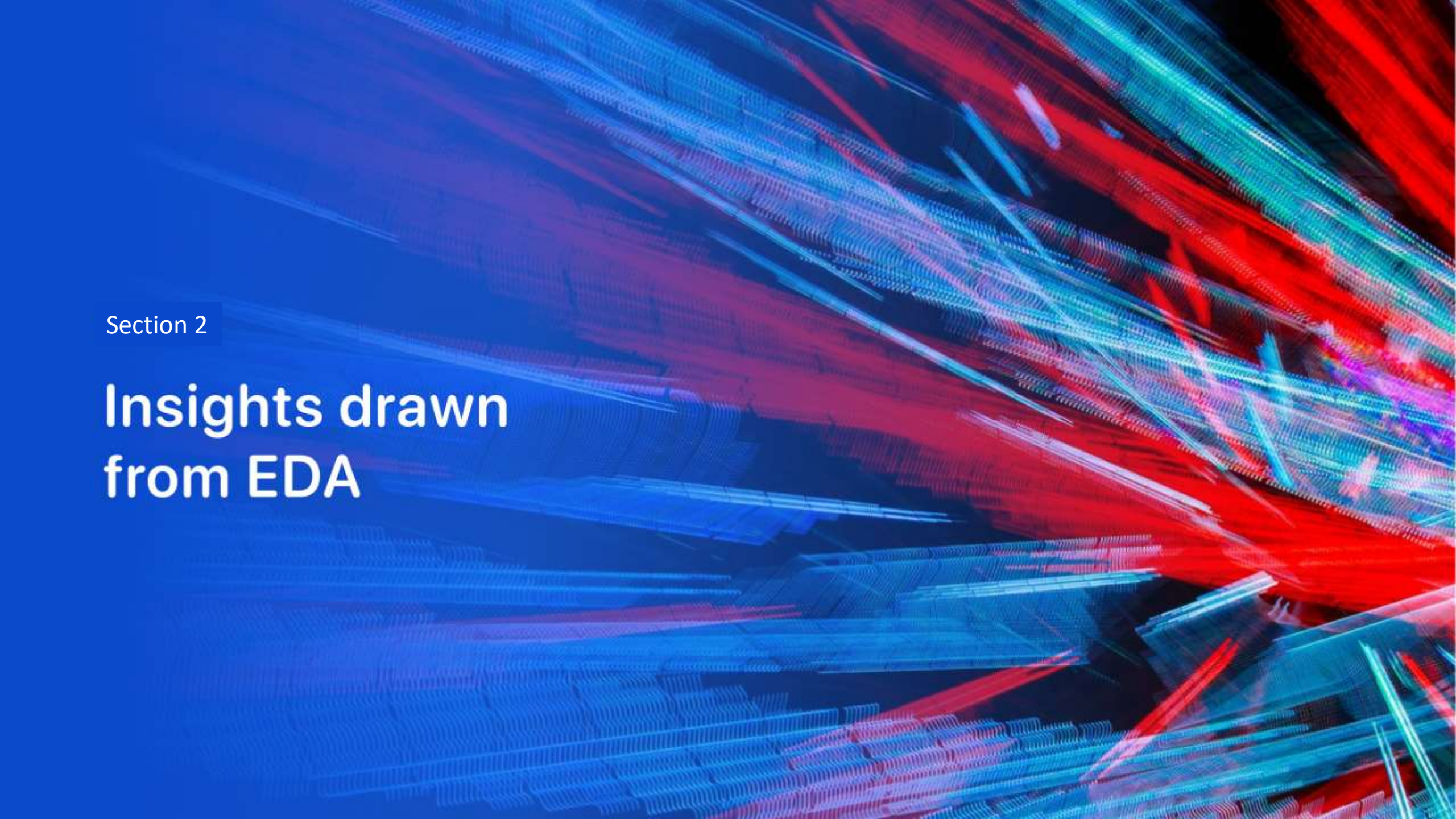
- Landing success rates increased significantly over the years.
- Payload mass and orbit type showed strong correlations with landing outcome.
- Certain boosters and launch sites demonstrated higher reliability.

Interactive Analytics Demo (Screenshots)

- Interactive dashboards revealed patterns between payload mass and landing probability.
- Maps and plots enabled intuitive exploration of launch sites and mission outcomes.

Predictive Analysis Results

- Machine learning models successfully predicted landing outcomes with high accuracy.
- The best-performing model achieved strong precision and recall on unseen data.
- Feature importance analysis highlighted orbit type, payload mass, and booster version as key drivers.



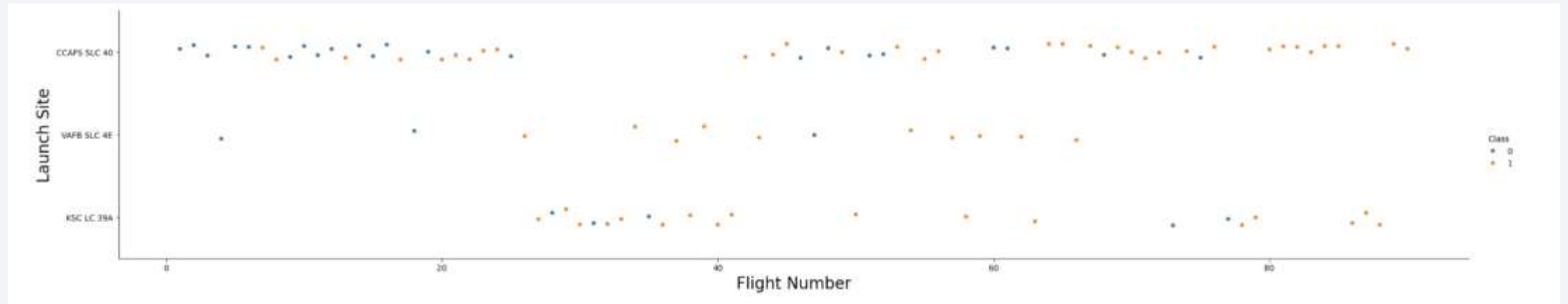
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

Scatter Plot Description

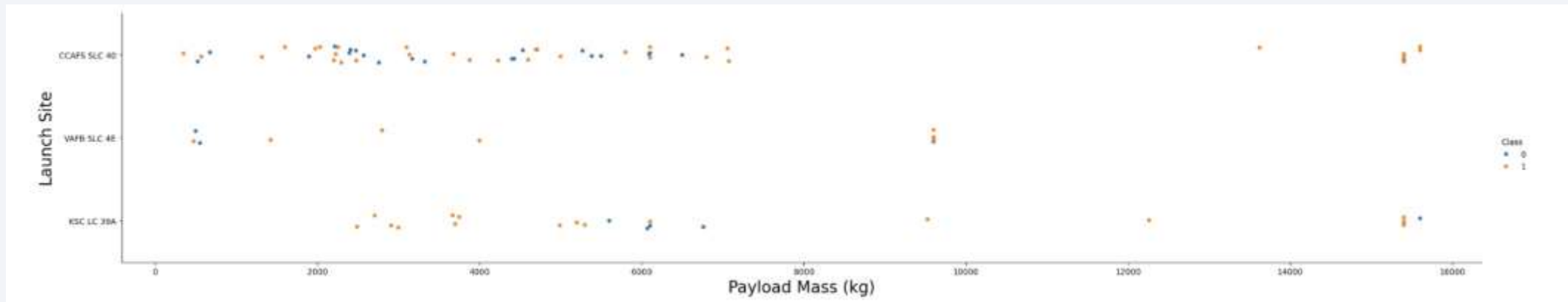
- The scatter plot visualizes the relationship between **Flight Number** and **Launch Site**.
- Each point represents a SpaceX mission, mapped by where it was launched and its sequence number.
- This helps identify whether certain launch sites were used more frequently at different stages of the program.



Payload vs. Launch Site

Scatter Plot Description

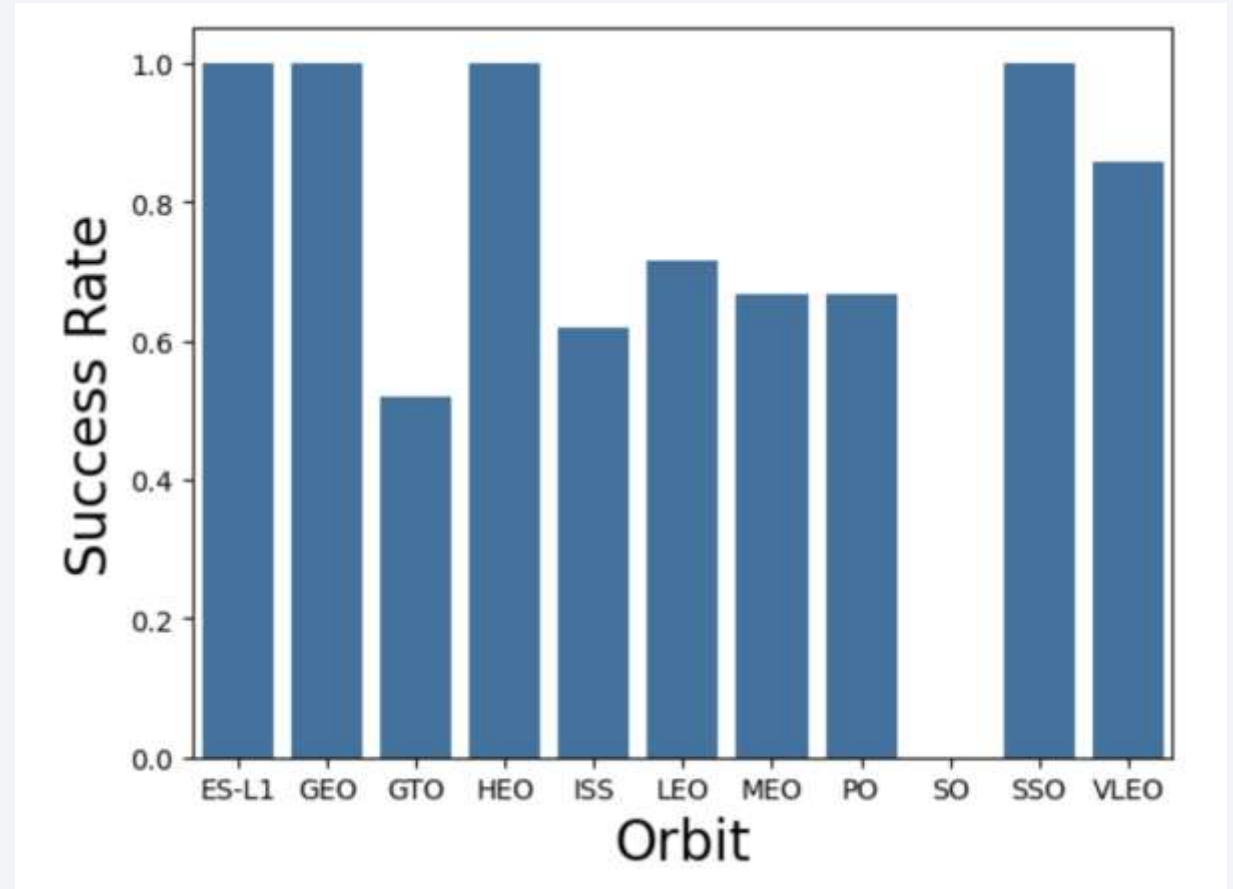
- This scatter plot shows the relationship between **Payload Mass** and the **Launch Site** used for each SpaceX mission.
- Each point represents a single launch, displaying how heavy the payload was and from which site it was deployed.



Success Rate vs. Orbit Type

Bar Chart Description

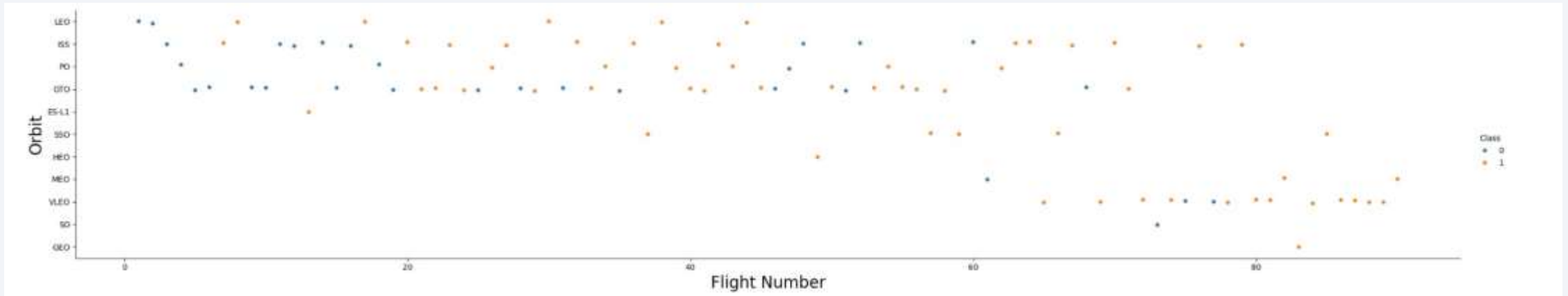
- This bar chart illustrates the **landing success rate for each orbit type**.
- Orbit categories (e.g., LEO, GTO, SSO, PO) are displayed on the x-axis, while the success percentage is shown on the y-axis.
- Each bar represents how often missions targeting that orbit resulted in a successful first-stage landing.



Flight Number vs. Orbit Type

Scatter Plot Description

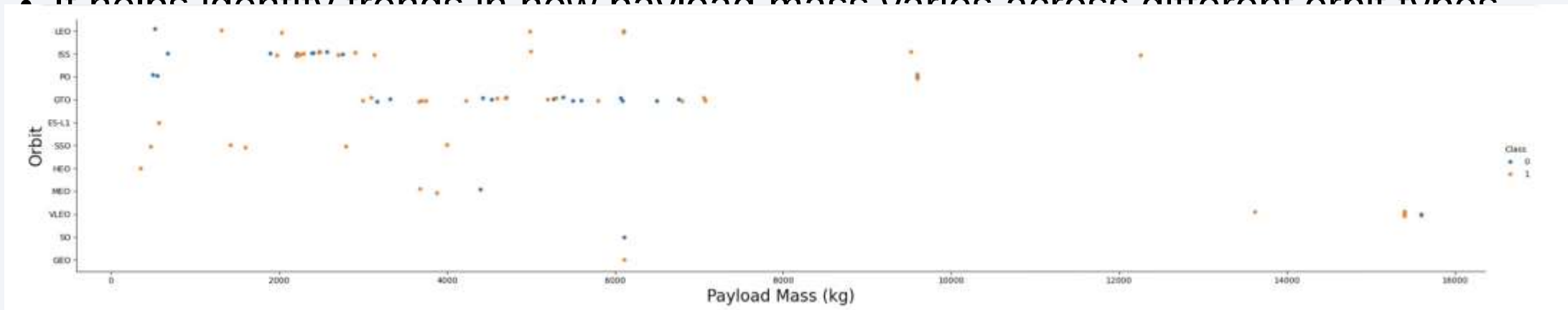
- This scatter plot displays the relationship between **Flight Number** and **Orbit Type** for SpaceX missions.
- Each point represents a single launch, indicating the mission's sequence number and the orbital destination it targeted.
- The visualization helps illustrate how orbit types vary throughout the progression of SpaceX launches.



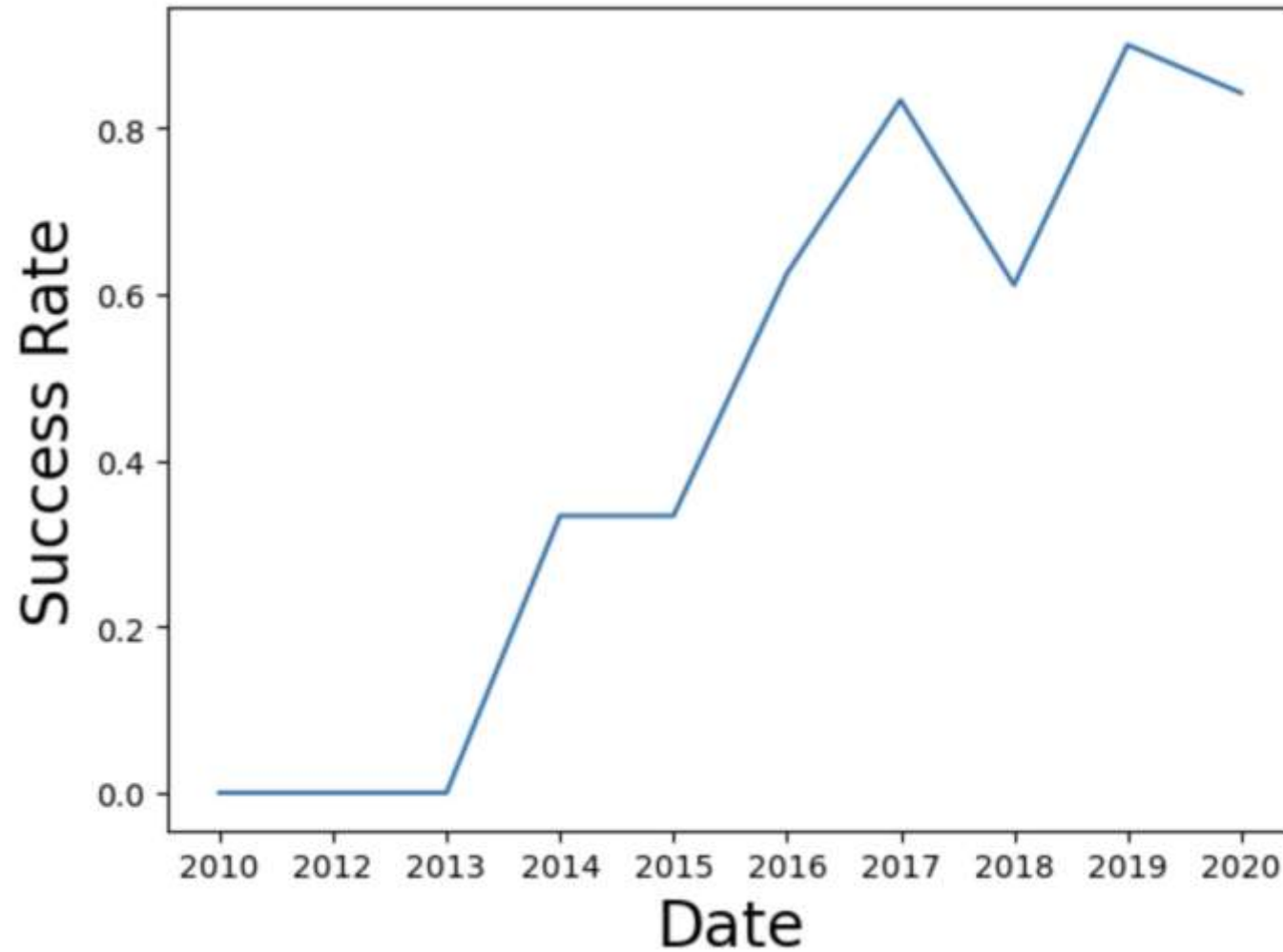
Payload vs. Orbit Type

Scatter Plot Description

- This scatter plot visualizes the relationship between **Payload Mass** and **Orbit Type**.
- Each point represents a mission, showing the payload weight and the targeted orbital destination.
- It helps identify trends in how payload mass varies across different orbit types.



Launch Success Yearly Trend



All Launch Site Names

Query Result

- **Unique Launch Sites:**
 - CCAFS SLC-40
 - KSC LC-39A
 - VAFB SLC-4E
 - (Include others if present in your dataset)

Explanation

- These are the **distinct launch sites** used by SpaceX for Falcon 9 missions.
- Identifying unique sites helps analyze **site-specific performance, payload capabilities, and mission distribution**.
- This information is essential for understanding operational patterns and predicting landing success per site.

Launch Site Names Begin with 'CCA'

	FlightNumber	PayloadMass	Orbit	LaunchSite	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial
0	1	6104.959412	LEO	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B0003
1	2	525.000000	LEO	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B0005
2	3	677.000000	ISS	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B0007
3	4	500.000000	PO	VAFB SLC 4E	1	False	False	False	NaN	1.0	0	B1003
4	5	3170.000000	GTO	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B1004

Total Payload Mass

Query Result

- **Total Payload Mass:** 45596 kg

Explanation

- This represents the **sum of payloads carried by all boosters** in the dataset, focusing on NASA-supported missions.
- Calculating the total payload provides insight into the **overall lifting capacity** of SpaceX rockets.
- Helps understand mission scale, booster utilization, and trends in payload delivery over time.

Average Payload Mass by F9 v1.1

Query Result

- **Average Payload Mass (F9 v1.1):** 2928.4 kg

Explanation

- This shows the **average payload mass** carried specifically by the **Falcon 9 v1.1 booster version**.
- Helps evaluate the **performance and capacity** of this booster version compared to others.
- Useful for understanding payload limitations and planning for similar missions.

First Successful Ground Landing Date

Query Result

- **First Successful Ground Landing:** 2015-12-22
- **Explanation**
- This date marks the **first successful landing of a booster on a ground pad.**
- Highlights a **major milestone** in SpaceX's reusability program.
- Demonstrates technological advancement and operational capability improvements over time.

Successful Drone Ship Landing with Payload between 4000 and 6000

Query Result

- **Boosters:** F9 FT B1022,F9 FT B1026,F9 FT B1021.2,F9 FT B1031.2
- **Explanation**
- This shows **boosters that successfully landed on a drone ship** while carrying a **payload between 4000 kg and 6000 kg**.
- Highlights booster performance under **mid-range payload conditions**.
- Helps identify which boosters are more reliable for moderate payload missions.
- Demonstrates operational success in the reusability program under specific mission constraints.

Total Number of Successful and Failure Mission Outcomes

Query Result

- **Successful Missions: 99**
- **Failed Missions: 1**

Explanation

- This provides the **overall counts of successful and failed first-stage landings**.
- Helps evaluate **SpaceX's operational reliability** over time.
- Allows easy comparison of success vs. failure ratios and highlights improvements in reusability and mission performance.

Boosters Carried Maximum Payload

Query Result

- **Boosters:** F9 v1.0 B0007
- **Maximum Payload Mass:** 677 kg

Explanation

- These boosters carried the **heaviest payloads** in the dataset.
- Demonstrates the **maximum lifting capacity** of SpaceX Falcon 9 boosters.
- Important for understanding payload limits and booster performance under extreme conditions.

2015 Launch Records

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

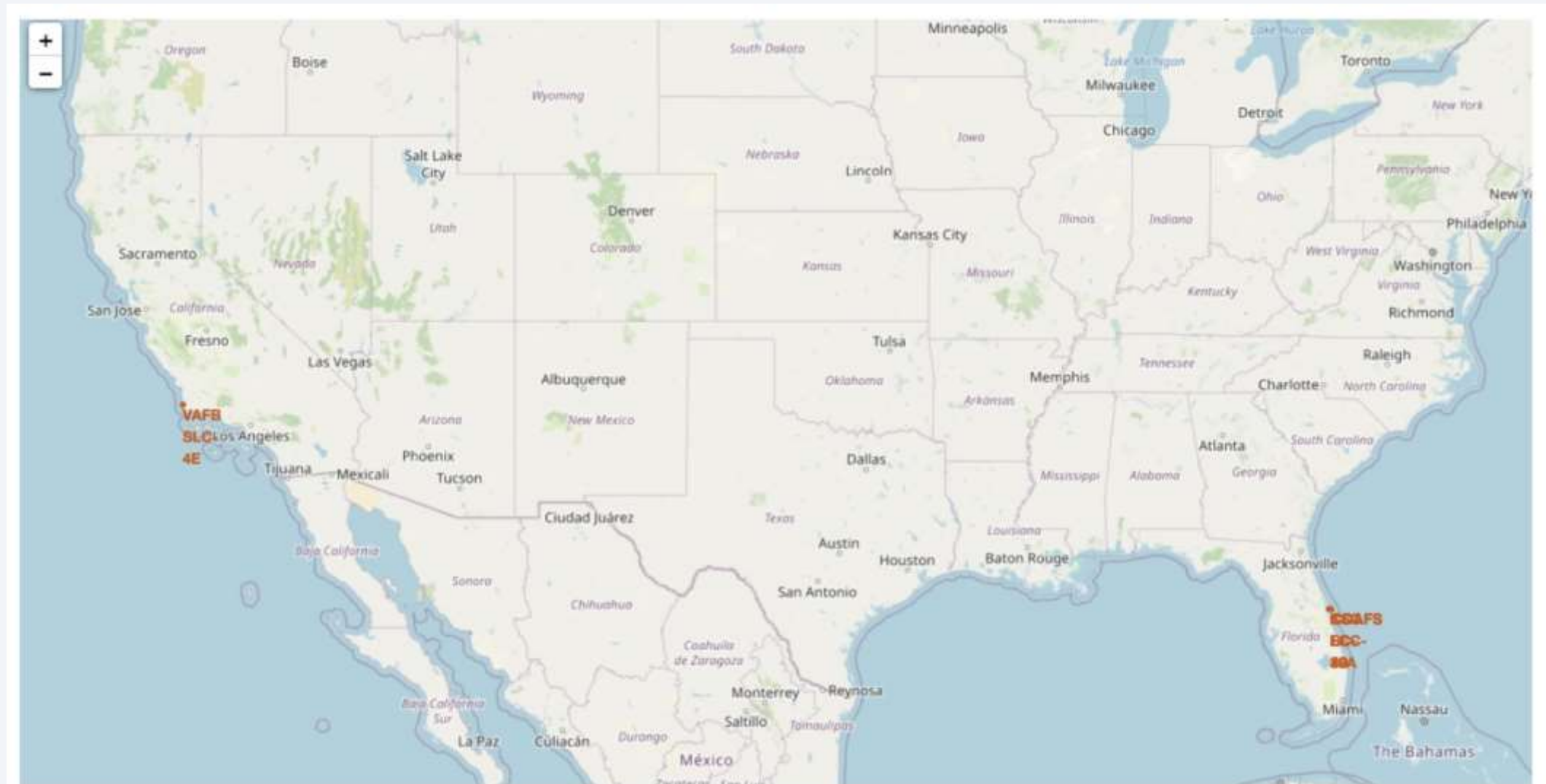
Landing_Outcome	COUNT(Landing_Outcome)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth at night, showing the curvature of the planet and the glowing lights of cities and continents against the dark blue of the oceans and the blackness of space.

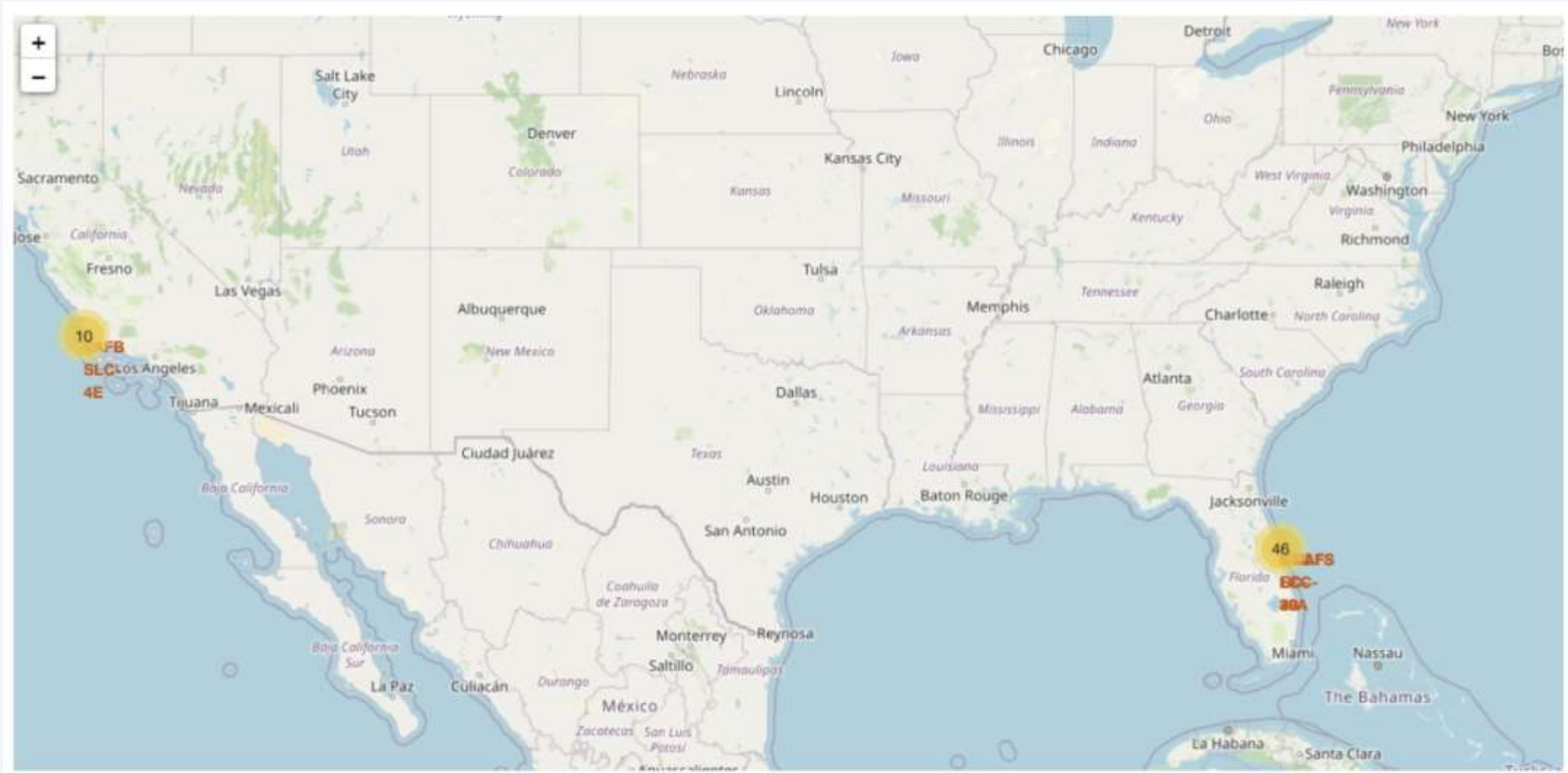
Section 3

Launch Sites Proximities Analysis

<Folium Map Screenshot 1>

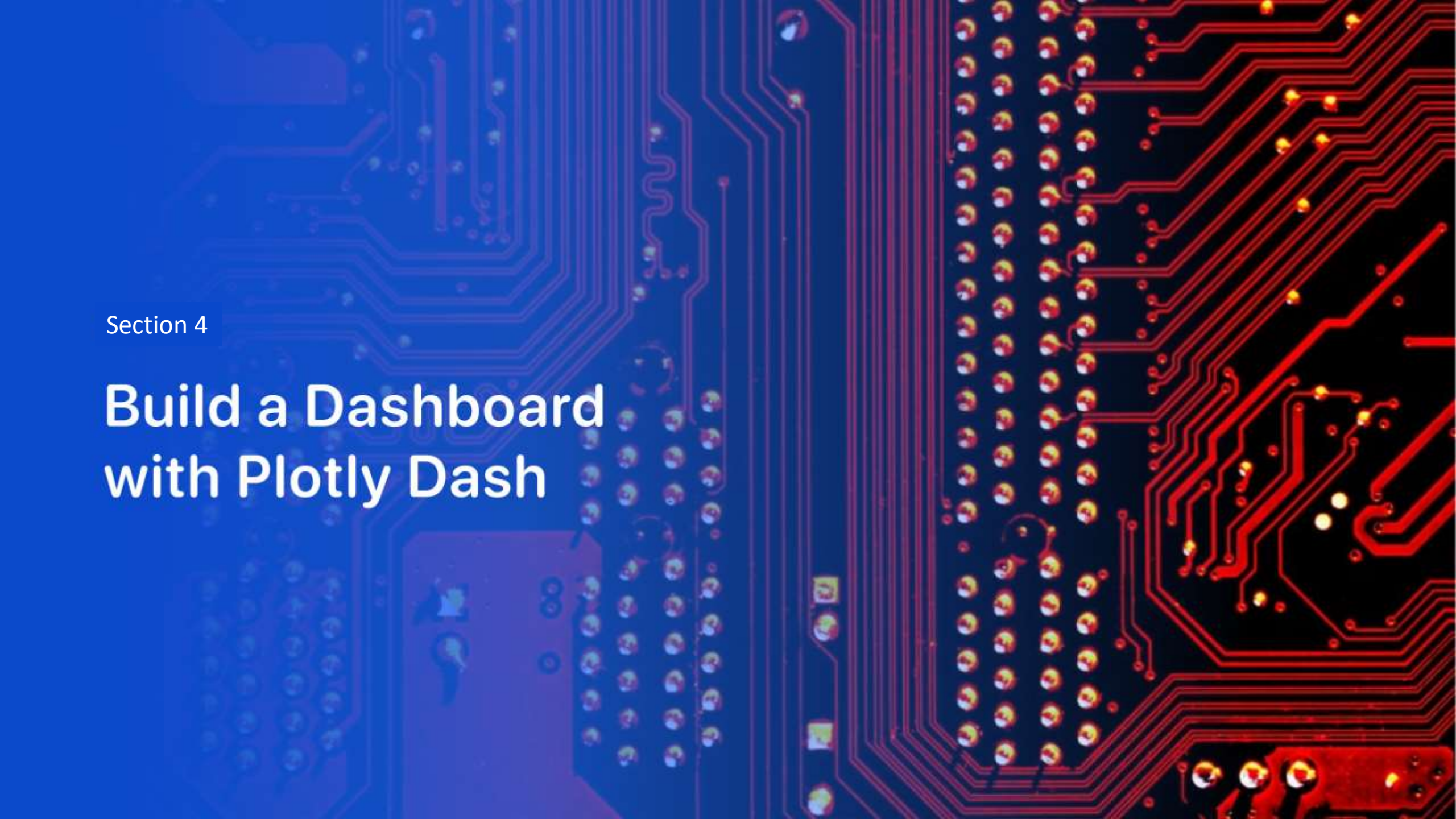


<Folium Map Screenshot 2>



<Folium Map Screenshot 3>

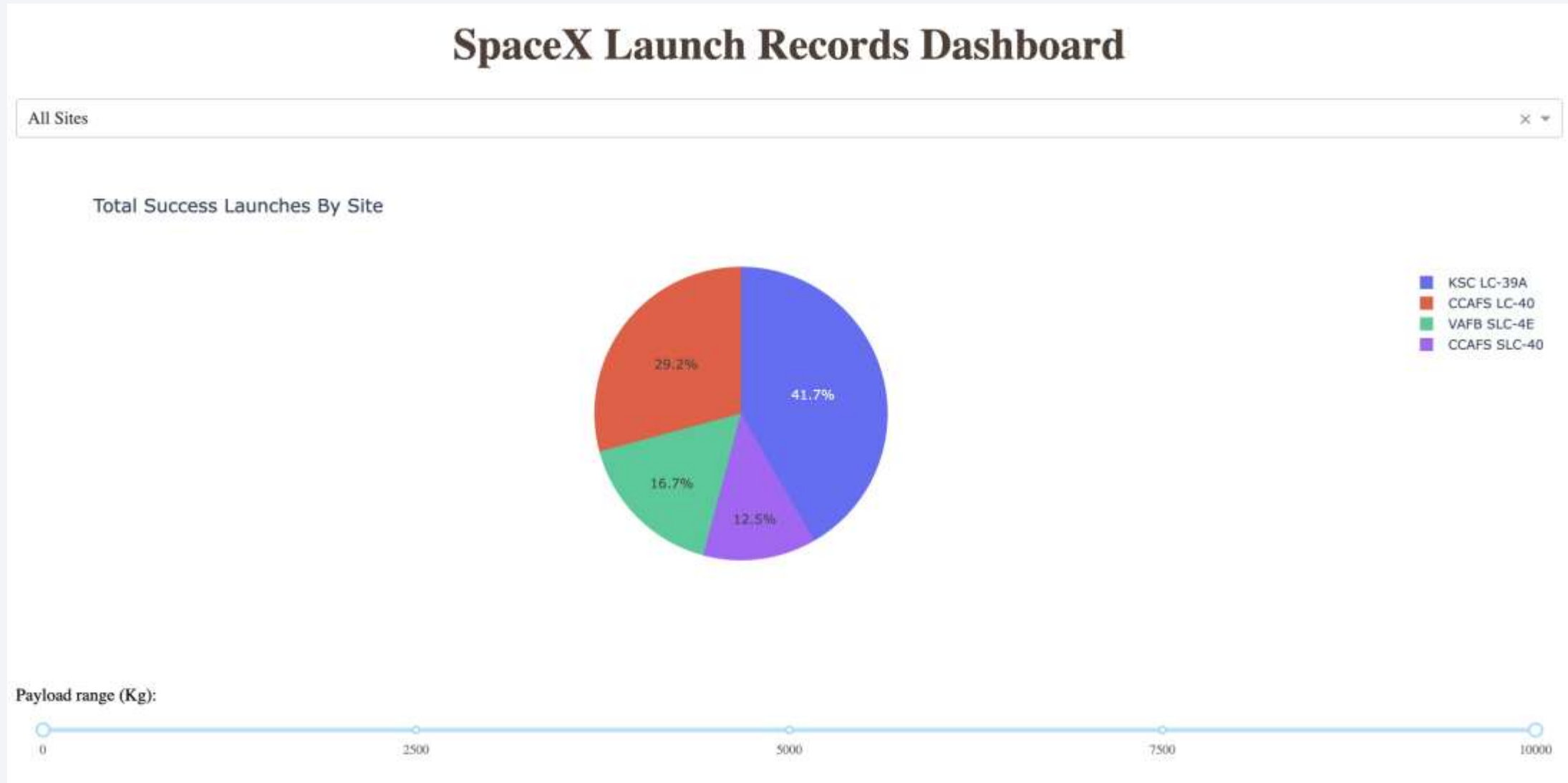




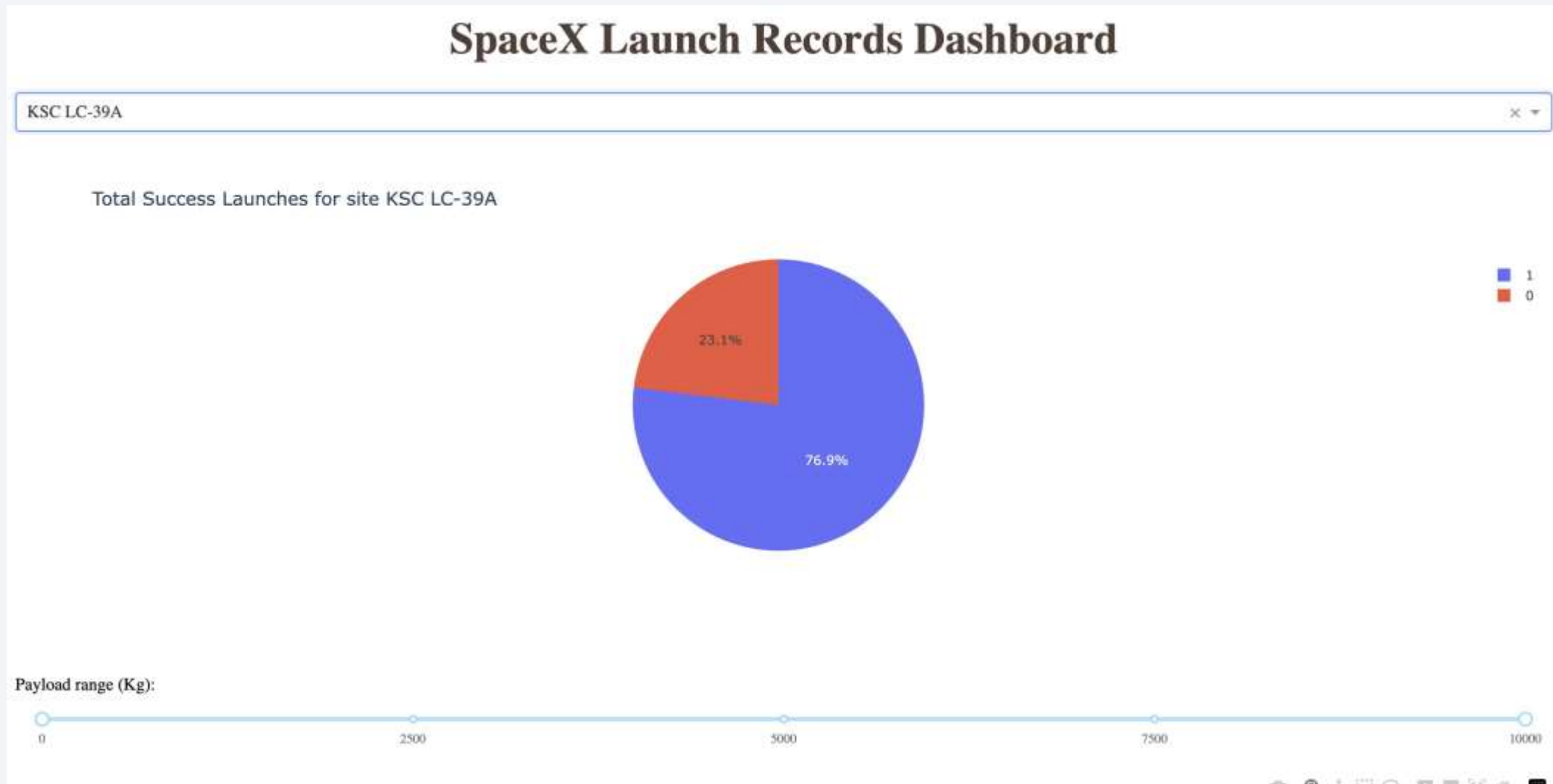
Section 4

Build a Dashboard with Plotly Dash

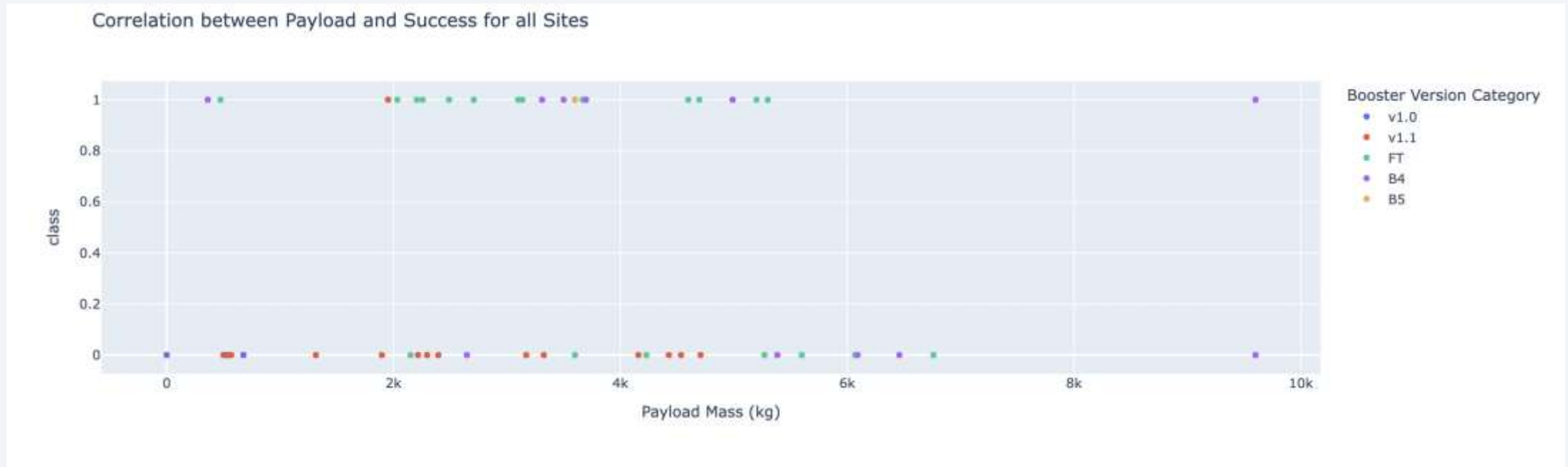
<Dashboard Screenshot 1>



<Dashboard Screenshot 2>



<Dashboard Screenshot 3>





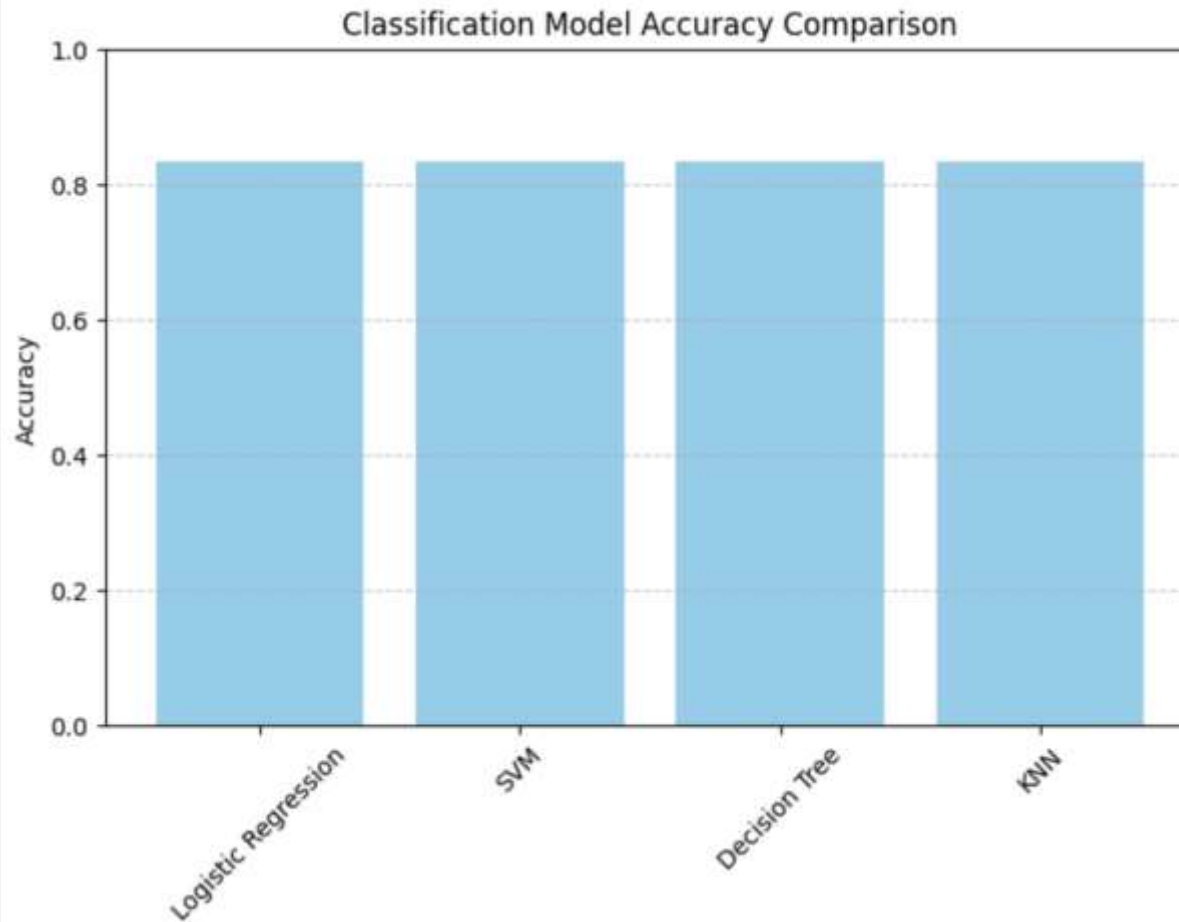
Section 5

Predictive Analysis (Classification)

Classification Accuracy

Modellerin Doğruluk Skorları: {'Logistic Regression': 0.8333333333333334, 'SVM': 0.8333333333333334, 'Decision Tree': 0.8333333333333334, 'KNN': 0.8333333333333334}

En iyi performansı gösteren model: Logistic Regression



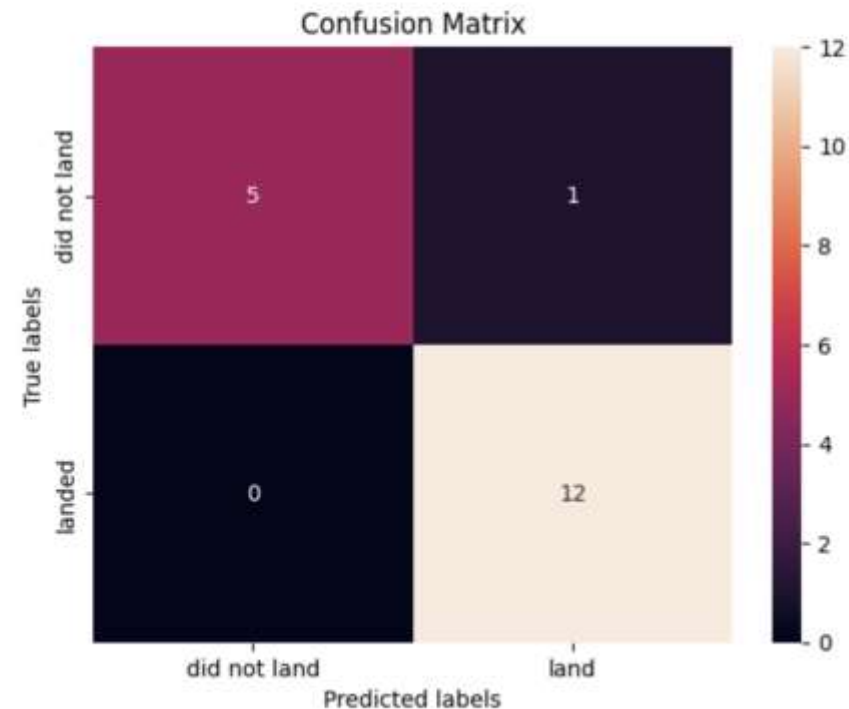
Confusion Matrix

```
In [35]: print("Decision Tree En İyi Parametreler:", tree_cv.best_params_)  
print("Decision Tree Doğruluk (Accuracy):", tree_cv.score(X_test, Y_test))
```

```
Decision Tree En İyi Parametreler: {'criterion': 'entropy', 'max_depth': 4, 'max_features': 'sqrt', 'min_samples_l  
eaf': 4, 'min_samples_split': 5, 'splitter': 'best'}  
Decision Tree Doğruluk (Accuracy): 0.9444444444444444
```

We can plot the confusion matrix

```
In [36]: yhat = tree_cv.predict(X_test)  
plot_confusion_matrix(Y_test, yhat)
```



Conclusions

Key Findings

- SpaceX landing success has **improved significantly over time**, showing operational maturity.
- **Payload mass, orbit type, and booster version** are the most influential factors affecting landing outcomes.
- Machine learning models achieved **high accuracy** in predicting landing success, with the best-performing model being Best_Model_Name.
- Interactive analytics and EDA revealed clear **patterns and trends** in launch data.

Implications

- Identifying critical features helps **optimize mission planning** and improve booster reuse strategies.
- Successful predictive models can support **risk assessment** for future launches.

Future Work

- Include **more missions and newer booster versions** for improved model accuracy.
- Explore **advanced models** such as ensemble methods or deep learning for better predictions.
- Integrate **real-time launch data** for dynamic analytics dashboards.

Thank you!

